

Julius Amankwah

Senior Backend Engineer & ML Systems

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PROFESSIONAL EXPERIENCE

Senior Software Engineer, Data Platforms | Meridian Health Systems | Portland, OR | March 2024 – Present

Led the design and launch of an LLM-powered clinical documentation assistant that automates medical note generation for over 800 physicians across Meridian's network. The system ingests patient encounter data via HL7 FHIR APIs, processes clinical context through a fine-tuned language model, and generates structured notes that integrate directly into the Epic EHR system. Implemented robust error handling and compliance validation layers to ensure HIPAA-compliant operation, reducing note completion time by approximately 45 minutes per physician per day while maintaining clinical accuracy standards. Collaborated with clinical informatics teams and physicians to define output requirements, iteratively refining prompts and model behavior based on production feedback from active users. Built comprehensive monitoring and audit trails for all model outputs to support regulatory review and continuous quality assurance. The system currently processes over 2,000 clinical encounters daily and has achieved 92% adoption among eligible providers, demonstrating genuine clinical utility rather than proof-of-concept value.

Staff Backend Engineer, Infrastructure | Cascade Data Solutions | Seattle, WA | June 2021 – February 2024

Owned the data ingestion and transformation pipeline that powered Cascade's core analytics platform, serving 150+ enterprise clients in life sciences, pharmaceutical, and healthcare sectors. Architected a highly available message-driven system using Kafka and Apache Flink to process 500+ million events daily from diverse clinical and research databases. Designed and maintained multiple microservices for data validation, deduplication, and enrichment, ensuring data quality standards that met stringent FDA and CLIA compliance requirements. Partnered closely with client success and medical affairs teams to troubleshoot data integration issues and configure custom transformation rules for specialized client workflows, often involving direct communication with research directors and data governance officers at partner institutions. Implemented observability infrastructure using Prometheus and Grafana, reducing mean time to resolution for data pipeline failures from 6 hours to under 30 minutes. Led the migration of critical services from on-premises Kubernetes clusters to AWS EKS, improving reliability metrics and reducing infrastructure costs by 35%.

Backend Engineer | Synapse Analytics | Boston, MA | July 2019 – May 2021

Developed and maintained REST APIs and backend services for Synapse's precision medicine platform, which aggregates genomic, phenotypic, and clinical data for research institutions. Built distributed data processing workflows using Python and Go to normalize and index patient cohort data across multiple healthcare networks. Collaborated with bioinformaticians, clinical geneticists, and research coordinators to translate complex domain requirements into scalable systems, often requiring deep understanding of genomic data formats and clinical trial protocols. Optimized database query patterns and implemented caching strategies that reduced API response latencies from 8 seconds to under 500 milliseconds for large cohort searches, directly enabling more responsive user interfaces for researchers. Managed cloud infrastructure on Google Cloud Platform, including Compute Engine instances, Cloud Storage, and BigQuery datasets. Mentored two junior engineers on production deployment practices and incident response procedures.

Software Engineer | Vertex Health Networks | Chicago, IL | September 2018 – June 2019

Contributed to the development of a patient engagement and outcomes tracking platform deployed across 12 regional health systems. Built backend services for secure patient data storage and retrieval, implemented secure API endpoints for mobile and web frontends, and supported integrations with multiple hospital information systems. Worked directly with clinical program managers and patient advocates to understand user needs and validate system behavior in real healthcare environments.

TECHNICAL SKILLS

Languages & Frameworks: Python, Go, JavaScript, SQL

Cloud & Infrastructure: AWS (EC2, RDS, S3, Lambda, ECS), Google Cloud Platform, Kubernetes, Docker

Data & ML: Apache Kafka, Flink, Airflow, LLM APIs (OpenAI, Anthropic), LangChain, prompt engineering, RAG systems

APIs & Integration: REST, gRPC, GraphQL, HL7 FHIR, DICOM, OAuth 2.0

Databases: PostgreSQL, MongoDB, Elasticsearch, BigQuery

Observability: Prometheus, Grafana, DataDog, structured logging

EDUCATION

BS Computer Science | University of Washington | Seattle, WA | 2018